Field Hours

CSCE 679 Data Visualization

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Marks, channels, and magnitude

You encode a table by choosing a mark and a channel. Marks are points, lines, and areas. Channels include position, length, area, and color. For a quantity comparison you spend position on a common scale first, then length. Area is a weaker magnitude channel, and hue is worse still. Values 10 and 30 should become bar lengths in the ratio 1 to 3. That ratio is the claim the reader must be able to see.

Radius is where area encodings go wrong. Area grows with the square of radius. If you set radii in the ratio 1 to 3 for the values 10 and 30, the areas stand in the ratio 1 to 9. You have overstated the larger value by a factor of 3. The mechanism is the square, not the reader's taste. Use a bar, or set areas, not radii, in the ratio 1 to 3.

Two encodings of 10 and 30.

Bar lengths

Lengths in ratio 1 to 3

The claim stays 10 and 30

Radii set to 1 and 3

Areas grow by the square

The picture says 1 to 9

Do not square the ratio

1. The values are 10 and 30. The true ratio is 1 to 3.

2. Bar lengths of 10 and 30 units keep that ratio.

3. Circle radii in the ratio 1 to 3 produce areas in the ratio 1 to 9. That overstates the larger value by a factor of 3.

Problem 1. Values are 10 and 30. You set circle radii in the ratio 1 to 3. What area ratio does the reader see, and by what factor is the larger value overstated?

Answer. Areas stand in the ratio 1 to 9, because area grows with the square of radius. The true value ratio is 1 to 3, so the larger value is overstated by a factor of 3.

Problem 2. You need the reader to compare 10 and 30. Which channel do you spend first, and what length ratio should the bars have?

Answer. Spend position on a common scale first, and if you use bars, spend length. The lengths should be in the ratio 1 to 3, matching the values.

Lecture 2 · 90 minutes

Match the channel to the data type

Expressiveness means the channel can express the facts in the data and no extra facts. A nominal field has no order. If you paint it with a luminance ramp from light to dark, you invent an order the field does not have. Use distinct hues for nominal categories, and keep the count near 8. An ordinal field may use a light-to-dark ramp. A quantitative field should use position or length before you reach for color.

Effectiveness means people read one channel more accurately than another for the task you set. Test a readout, not a preference. Two bars of length 10 and 30 are easier to order than two hues standing for the same numbers. A rainbow scale adds a separate failure: hue is circular, so the ends are not an order. For one continuous measure, vary lightness of a single hue. That keeps the order inside 1 channel.

Type of data, then channel.

Nominal data get distinct hues

Keep the hue count near 8

Ordinal data may use lightness

Quantities get position or length

A department code is not a magnitude

1. The field is a department code with 6 values and no order.

2. A dark-to-light ramp would rank the codes. That ranking is not in the data, so the channel is not expressive.

3. You assign 6 distinct hues. Six is near the limit of about 8, and you do not use those hues to encode the budget.

Problem 1. A department code has no order. Why does a light-to-dark ramp fail expressiveness?

Answer. A ramp states an order. The codes do not have one, so the channel adds a fact the data do not contain. Use distinct hues instead, and keep the count near 8.

Problem 2. Why is a rainbow a bad encoding of one continuous measure, and what do you use instead?

Answer. Hue is circular, so the two ends are not an order. Vary the lightness of a single hue. That keeps the order inside 1 channel. Position or length is still the stronger magnitude channel if you have an axis free.

Lecture 3 · 90 minutes

Tables, trees, and networks

Arrangement encodes relationship by proximity. You keep values that the reader must compare in a shared frame. A network can be a node-link drawing or an adjacency matrix. On 8 nodes the maximum number of undirected edges is 28. At 20 edges a node-link drawing crosses a lot, and a matrix of 8 by 8 cells still shows

each edge as one mark.

Order inside the arrangement is part of the encoding. If you sort heatmap rows by a quantity, adjacency in the picture means adjacency in that quantity. If you leave the rows in the order of a file dump, adjacency means nothing and the reader will still try to read it. A tree laid out with the root at the top and 4 leaves in a row states depth by vertical position.

Twenty edges on eight nodes.

Node-link drawing

Crossings pile up at 20 edges

Better for a short path

Adjacency matrix

8 by 8 cells

Better for looking up a pair

Choose a matrix at 20 edges

1. You have 8 nodes. A complete undirected graph would have 28 edges. You have 20.

2. A node-link drawing of those 20 edges crosses heavily.

3. An 8 by 8 matrix shows each edge as a mark in a cell. You pick the matrix when the task is lookup of a pair.

Problem 1. How many undirected edges can 8 nodes have at most, and why is a node-link drawing a weak default at 20 edges?

Answer. The maximum is 28, from 8 times 7 divided by 2. At 20 edges the drawing crosses a lot. A matrix of 8 by 8 cells still shows every edge.

Problem 2. You leave heatmap rows in file order. What does vertical adjacency mean, and what does it mean after you sort by a quantity?

Answer. In file order, adjacency means nothing, but readers will still try to read it. After you sort by a quantity, adjacency in the picture means adjacency in that quantity.

Lecture 4 · 90 minutes

What the eye does first

Some channels pop out before a deliberate search. A single red point among gray points is found in a time that barely grows with the number of gray points. A single point defined only by a conjunction, red and square, among red circles and gray squares, forces a serial search. You use one pop-out channel for the thing the reader must find first. A dashboard with 12 bright colors has spent the pop-out and left the reader to scan.

Contrast is local. The same gray looks lighter on black than on white, so a cell's neighbors change how you read a heatmap. Separate regions when the reader must match a cell to a legend. A small absolute gap is also easier on a short bar than on a long one: adding 1 to a bar of length 5 changes it by a fifth, and adding 1 to a bar of length

50 changes it by a fiftieth.

Pop-out against a conjunction.

One red point

Color differs alone

Search stays fast

Red and square together

Two features define the target

Search becomes serial

A difference of 1 on two bars

1. Bar A has length 5. Adding 1 makes length 6, which is a change of 1/5.

2. Bar B has length 50. Adding 1 makes length 51, which is a change of 1/50.

3. If that gap of 1 is the claim, encode the difference directly. Do not hope the reader sees it on the long bar.

Problem 1. Why can a reader find one red point among gray points faster than a point that is red and square among red circles and gray squares?

Answer. A single channel, color, pops out, and the time barely grows with the number of gray points. The red square is a conjunction, so the reader has to search serially. You spend one pop-out channel on the fact that must be found first.

Problem 2. You add 1 to a bar of length 5 and 1 to a bar of length 50. Which change is the larger fraction, and what should you encode if the gap of 1 is the claim?

Answer. The change on the short bar is 1/5. The change on the long bar is 1/50. Encode the difference itself, not only the raw totals, when that gap is what the reader must see.

Lecture 5 · 90 minutes

Filter, select, and details

Interaction changes the subset or the detail without a new encoding design. Filter, select, and details on demand are the three mechanisms to build first. A filter keeps rows that pass a predicate, such as year = 2020, and draws only those. A brush on one chart selects marks, and a linked chart highlights the same ids. Details on demand print the raw values for 1 mark you point at.

Linking is what makes two views one tool. If a click in a list does not highlight the matching point, you have two separate pictures. A filter that drops 90 of 100 rows must say that 10 remain, and a control must restore the 90. Otherwise absence looks like a property of the data. Keep the filter animation short, a few tenths of a second, so a second click is not triggered by doubt.

One filter, with a way back.

1. Start from all 100 rows

2. Keep the rows for 2020

3. Show that 10 remain

4. Offer a control that restores 90

A brush that really links

1. The scatter and the list share document ids. The reader brushes 4 points.

2. The list highlights those same 4 ids. A click that does not highlight has not linked the views.

3. Details on demand show the raw numbers for 1 of the 4. The other marks stay unlabeled.

Problem 1. A filter hides 90 of 100 rows. What must the view say, and what control do you owe the reader?

Answer. It must say that 10 rows remain. A control must restore the 90. Otherwise the missing rows look like a property of the data rather than a view state.

Problem 2. Why is a list beside a scatter not yet one tool when a click in the list leaves the scatter unchanged?

Answer. Linking is the mechanism that makes two views one tool. Without the highlight on the matching point, you have two pictures that only share a topic.

Lecture 6 · 90 minutes

Sequence and annotation

A narrative is an ordered argument made of views. You choose which comparison the reader meets first. Change one thing per step: the filter, the encoding, or the annotation. With 3 outliers, use 3 steps or one step with 3 callouts, rather than one unlabeled scatter of 200 points. A title can state the claim, for example that region 2 fell after week 4. A title that only names the chart type does not carry that claim.

The reader of an interactive narrative can leave your order. That does not cancel the order. It means you also provide a scrubber or a step control with the count of steps visible, such as step 2 of 5. A reader who jumps to step 5 must still see the annotations that make step 5 true, not a chart that depended on a sentence you spoke over step 1. Put the claim on the view.

One change at each step.

State the claim in the title -> Change only the filter -> Annotate region 2 -> Leave the other encodings fixed

Three outliers, not one cloud

1. The scatter has 200 points and 3 outliers you need the reader to see.

2. You do not dump all 200 unlabeled. You either use 3 steps or one view with 3 callouts.

3. The title says what happened, not the chart type. A reader who opens step 2 of 5 still sees that

sentence on the view.

Problem 1. You have 3 outliers in a scatter of 200 points. How do you show them without changing the filter, the encoding, and the annotation in one step?

Answer. Change one thing per step. Use 3 steps, or one step with 3 callouts. Do not bury the outliers in a single unlabeled scatter, and do not retitle the view with only the chart type.

Problem 2. A reader jumps to step 5 of 5. Why must the claim still be written on that view?

Answer. The reader can leave your order. Step 5 cannot depend on a sentence you spoke over step 1. Put the claim on the view, and show the step count, such as step 2 of 5, on the control.

Lecture 7 · 90 minutes

When a third spatial axis helps

Use a third spatial dimension when the data are spatial. A terrain or a molecule already has height and depth the reader can interpret as space. You rotate it so the face that matters points at the camera, and occlusion is a fact about the object. Extruding 5 bars makes a front bar hide a back bar, and perspective makes two equal heights look different. You spent position on a depth the data do not have.

A 3D scatter is a different case from a terrain. With 40 points, some lie behind others, and a still image cannot be rotated. Prefer position for 2 attributes and lightness for the third. If you keep the 3D scatter, add a rotation the reader controls and a readout of all 3 coordinates of the point they select. A hidden point plus no readout is a missing value you introduced.

Depth that is real, and depth that is not.

A molecule

The third axis is space

Rotation resolves occlusion

Five extruded bars

The third axis is fake

A front bar hides a back bar

Forty points and a hidden one

1. The scatter has 40 points in 3 coordinates. In a still image, some points sit behind others.

2. You add a rotation the reader controls, plus a readout of all 3 coordinates of the selected point.

3. Without that readout, a hidden point is a missing value you created. Two attributes on position and the third on lightness avoid that hole.

Problem 1. Why may you use a third spatial axis for a molecule, and why not for 5 bar values?

Answer. A molecule already has spatial height and depth, and rotation treats occlusion as a fact about the object. Extruding 5 bars spends position on a depth the data do not have, hides a back bar, and

makes equal heights look different under perspective.

Problem 2. A still 3D scatter has 40 points. What two additions keep a hidden point from becoming a missing value you introduced?

Answer. A rotation the reader controls, and a readout of all 3 coordinates of the selected point. Otherwise the hidden point has no value the reader can recover.

Lecture 8 · 90 minutes

Defend an encoding

You can now choose a mark and a channel, refuse a channel that invents order, and refuse a radius that squares a ratio. You can choose a matrix for dense edges, keep 3D for spatial data, and report how many rows a filter leaves. You can order a few narrative steps so each step changes one thing. A reader should recover a ratio you encoded, such as 1 to 3, and tell a filter from a fact.

You will see designs fail in combinations. A 3D bar chart of a nominal field, colored with a rainbow and filtered with no count of remaining rows, has stacked 4 mistakes. The fix uses the same stack in reverse. Put the quantity on bar length in 2D, color the nominal field with a few distinct hues, and show that the filter kept 10 of 100 rows. Add one annotation that states the claim.

Undo four stacked mistakes.

Leave 3D for spatial data

Put the quantity on bar length

Paint the nominal field with hues

Say the filter kept 10 of 100

Rebuild one bad chart

1. The bad chart extrudes bars, paints a department code with a rainbow, and filters without saying how many rows remain.

2. You drop the extrusion, map the quantity to length in the ratio the data have, and give the code distinct hues.

3. The filter reports 10 of 100. One annotation states the claim. A reader can recover the ratio and can tell the filter from the data.

Problem 1. A chart encodes 10 and 30 as circle radii in the ratio 1 to 3, and it paints an unordered code with a light-to-dark ramp. What two failures are those, and what do you change?

Answer. The radii square the ratio: areas become 1 to 9, overstating the larger value by a factor of 3. Use lengths in the ratio 1 to 3. The ramp invents an order the code does not have. Use distinct hues, near a count of 8 at most.

Problem 2. Your fix is 2D bars, distinct hues for a nominal field, and a filter that keeps 10 of 100 rows, plus one claim in the title. Which earlier mechanisms did you combine, and what readout shows the filter is not a fact about the data?

Answer. You combined the length channel, an expressive nominal encoding, and a filter that reports its own state. The readout is the count: 10 remain and 90 were removed by the view. Without that count, absence looks like a property of the data.